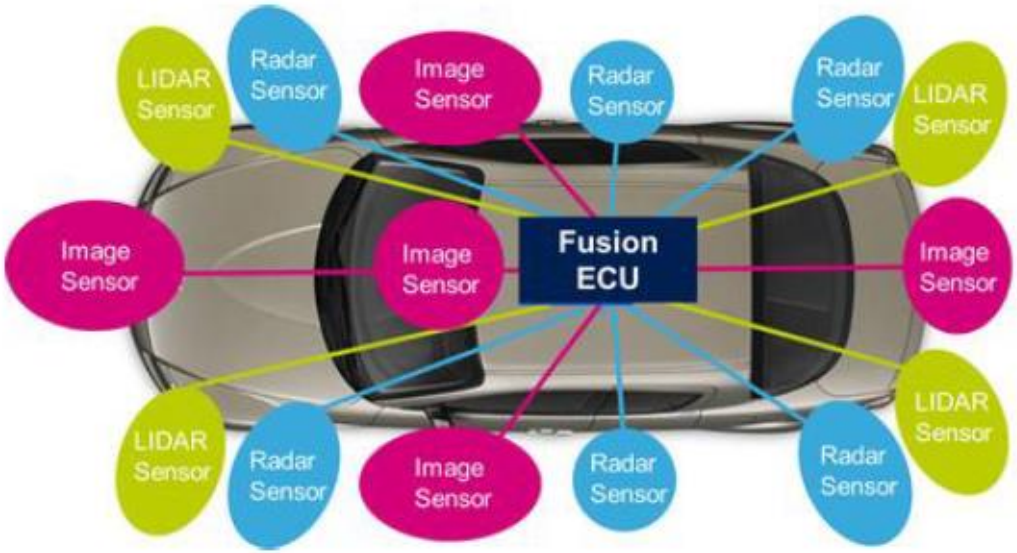
Sensors Used in Autonomous Driving

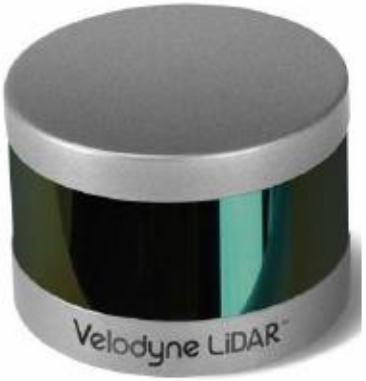
Sparsh Mittal

Sensors



Camera

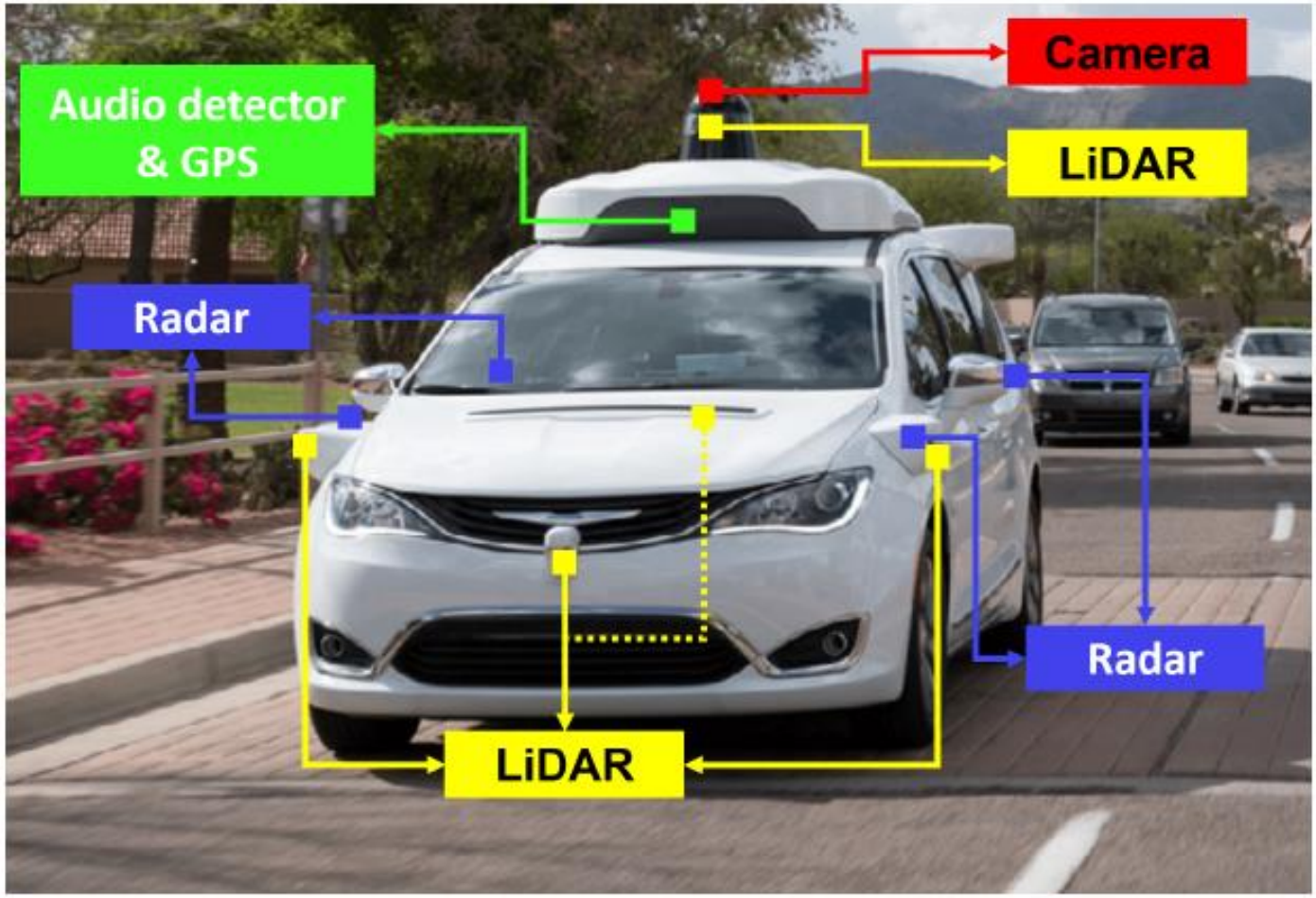
Radar



LIDAR

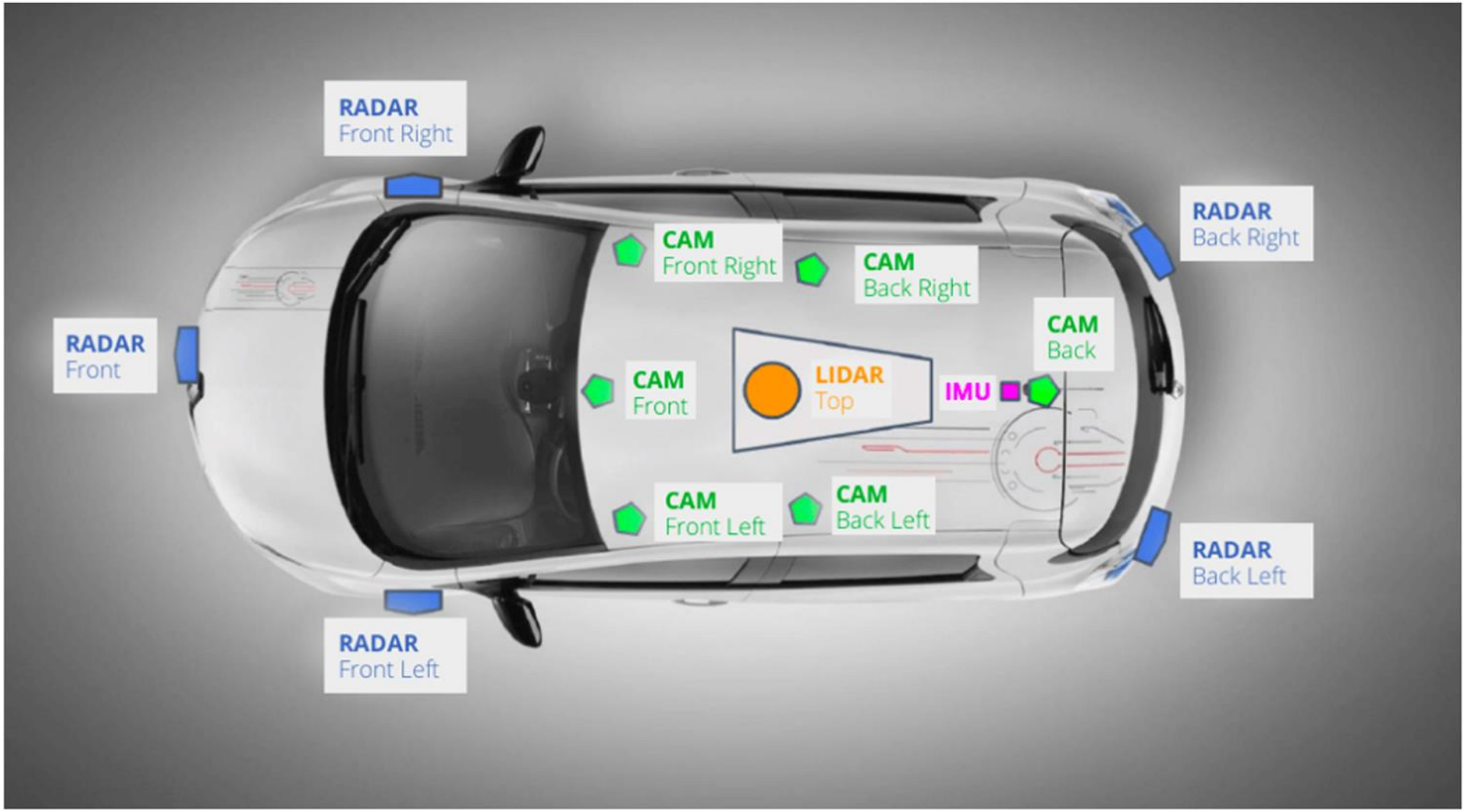
Courtesy: MIT "Deep Learning for Self-Driving Cars" course

Sensor placement



LIDAR: Light Detection and Ranging

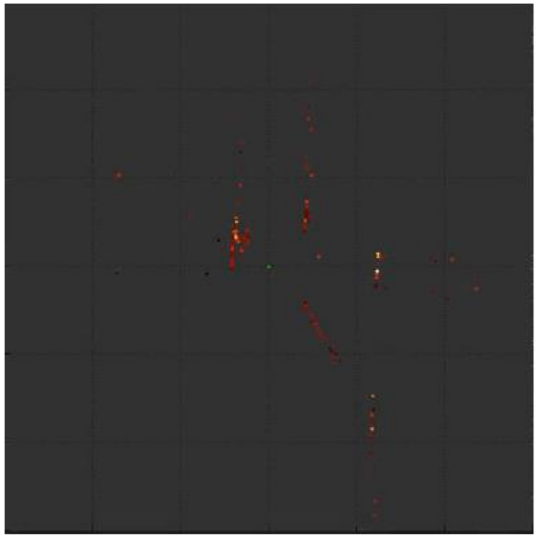
Sensors used in nuTonomy self-driving car



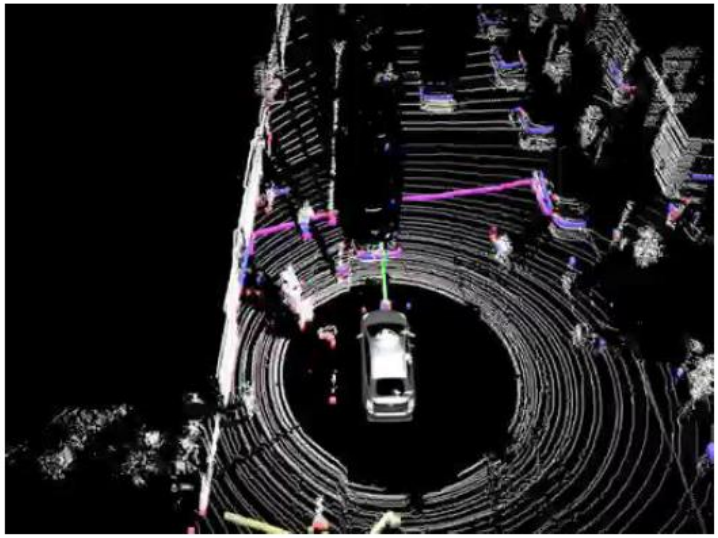
Comparison of Radar and lidar

- Radar: Cheap, low-resolution, used most widely
- works well in extreme weather

- Lidar: Expensive, much higher resolution than radar, extremely accurate depth info, 360 degree visibility



Radar



Lidar

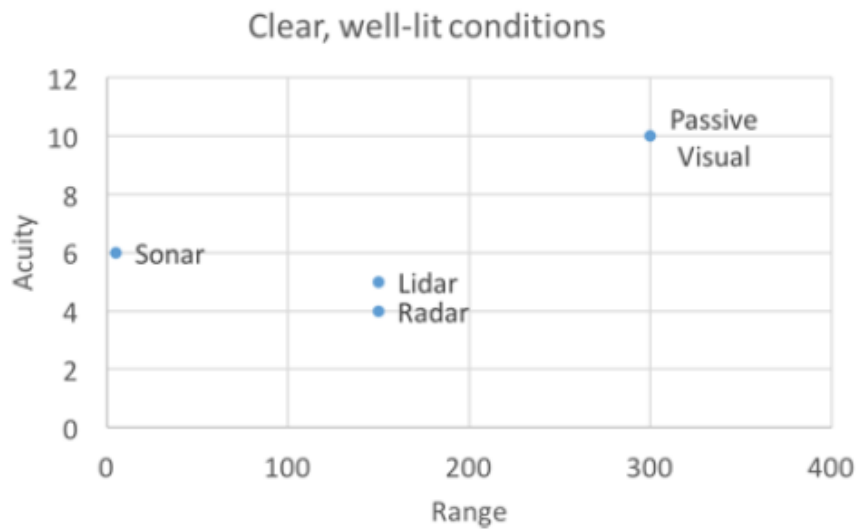
Courtesy: MIT "Deep Learning for Self-Driving Cars" course

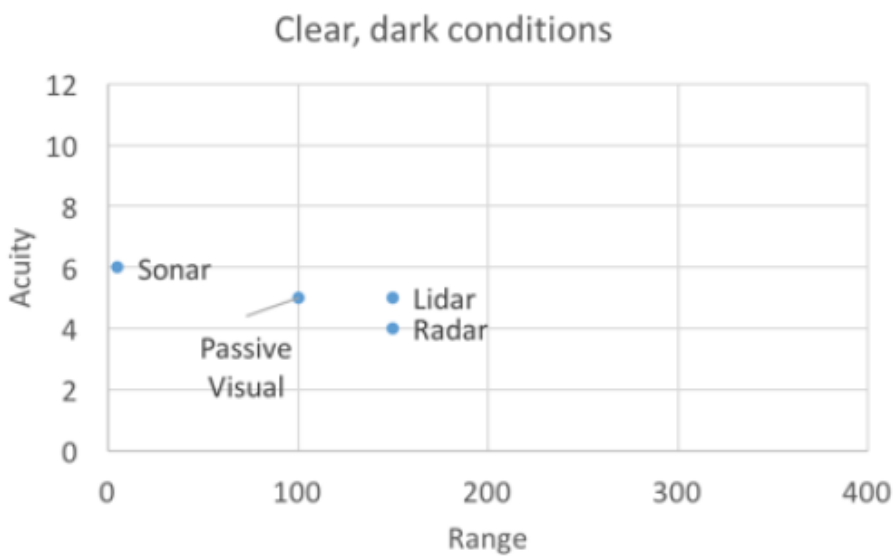
Camera: pros and cons

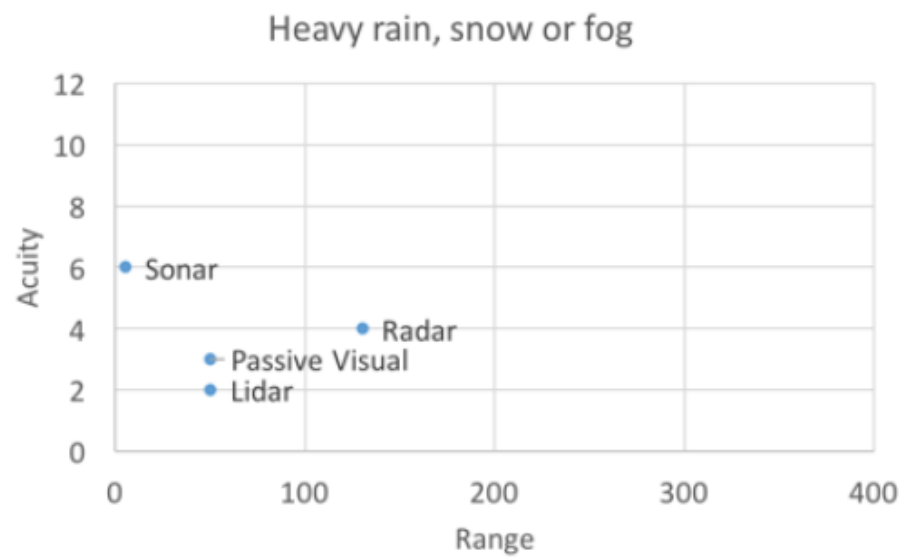
- Camera: cheap, provide huge data, so deep learning is required.
- Humans use computer vision to drive
- Poor at depth estimation
- Not good in extreme weather

Courtesy: MIT “Deep Learning for Self-Driving Cars” course

Comparison of sensors

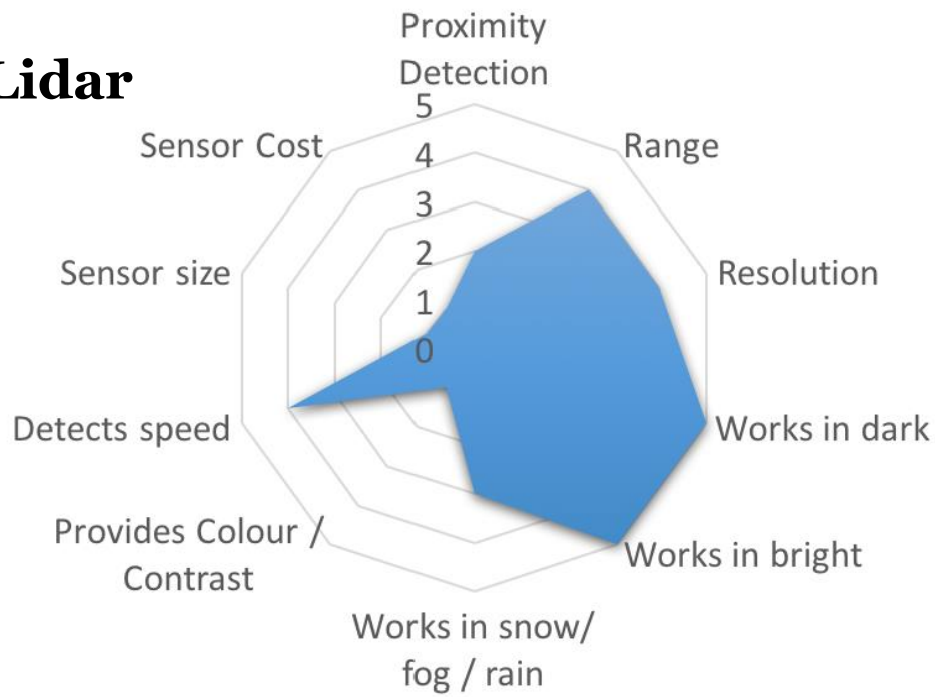




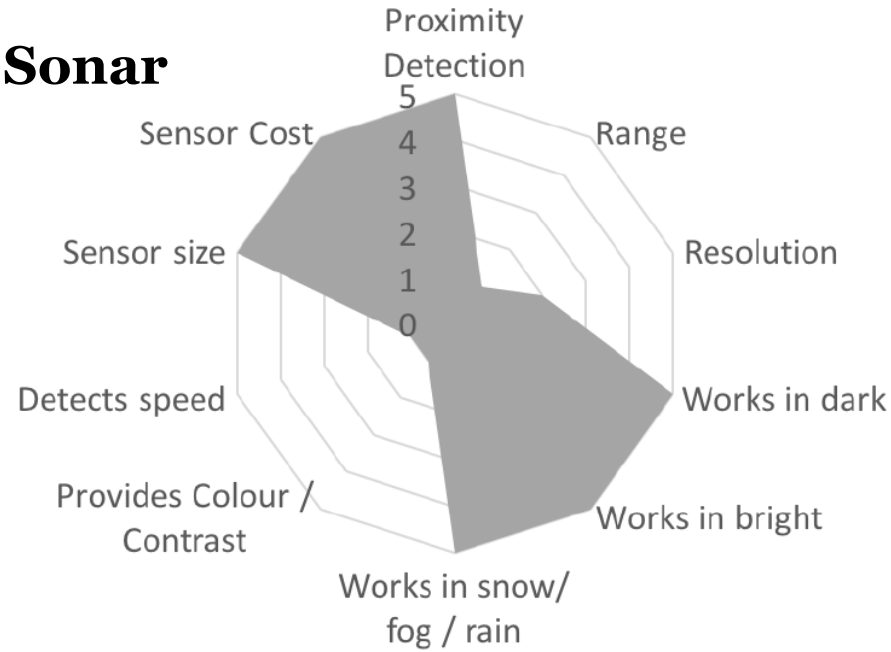


Courtesy: MIT “Deep Learning for Self-Driving Cars” course

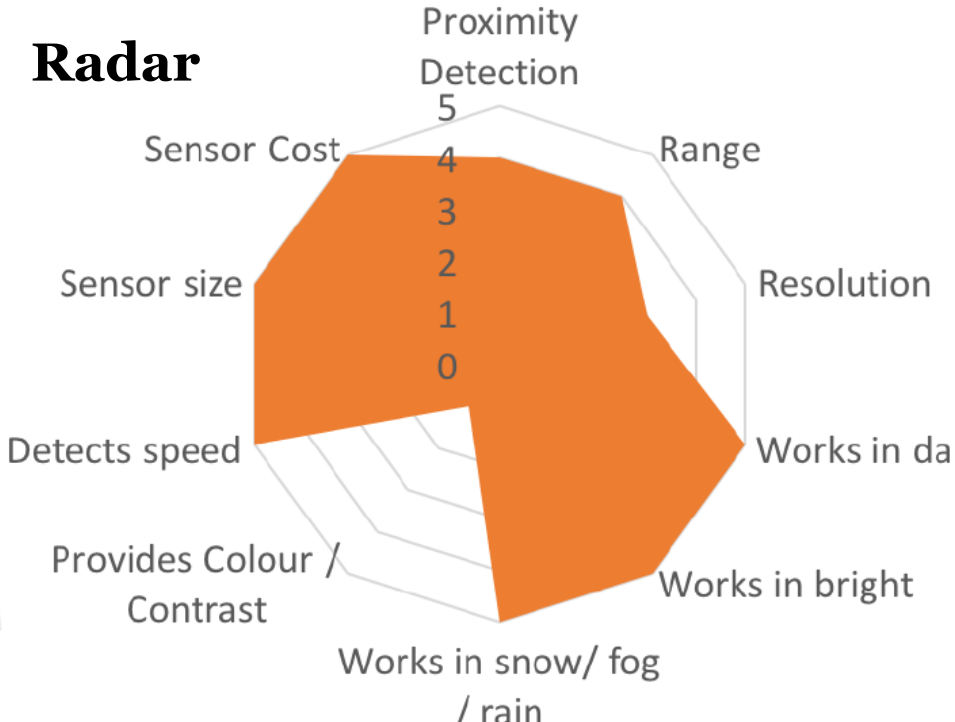
Lidar



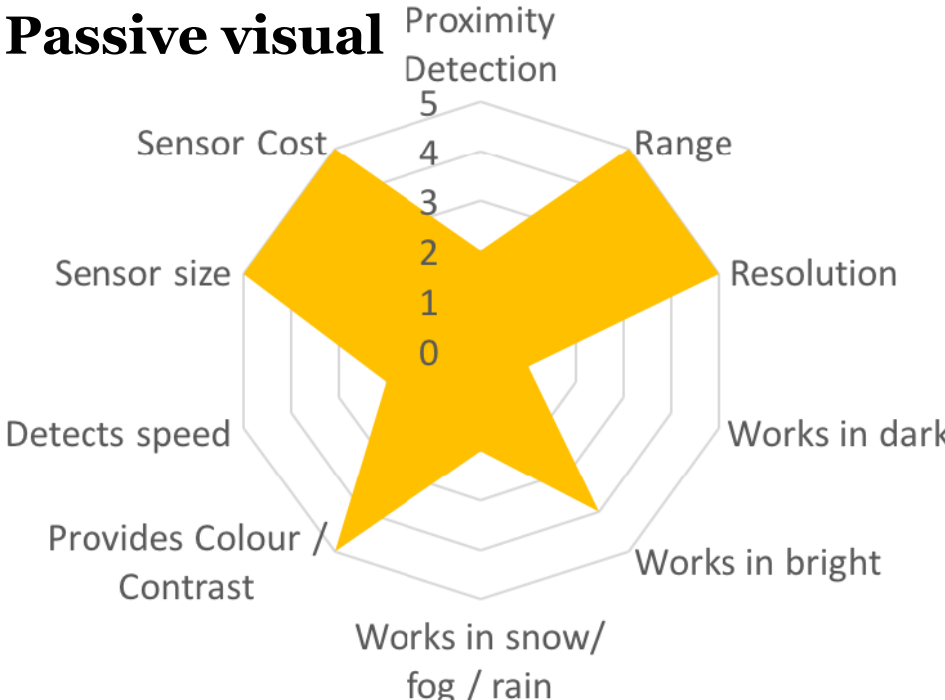
Sonar



Radar



Passive visual



State-of-art in ADVs in leading companies

Manufacturer	Mobileye [47]	Tesla [15, 70]	Nvidia/Audi [67]	Waymo [22, 68, 77]
Automation	level 2	level 2	level 3	level 3
Platform	SoCs	SoCs + GPUs	SoCs + GPUs	SoCs + GPUs
Sensor	camera	camera, radar	lidar, camera, radar	lidar, camera, radar

Challenges of LIDAR

- LIDAR: a remote sensing device used to examine surroundings of the vehicle at high precision by sending light beams
- **Benefits:** High precision
- **Challenge:** Very high cost, e.g., US$ 75,000, which is costlier than the car.
- => Avoid using LIDAR. Instead, use vision-based ADVs using cameras and radars which are much cheaper. This is the focus of companies like Tesla and Mobileye